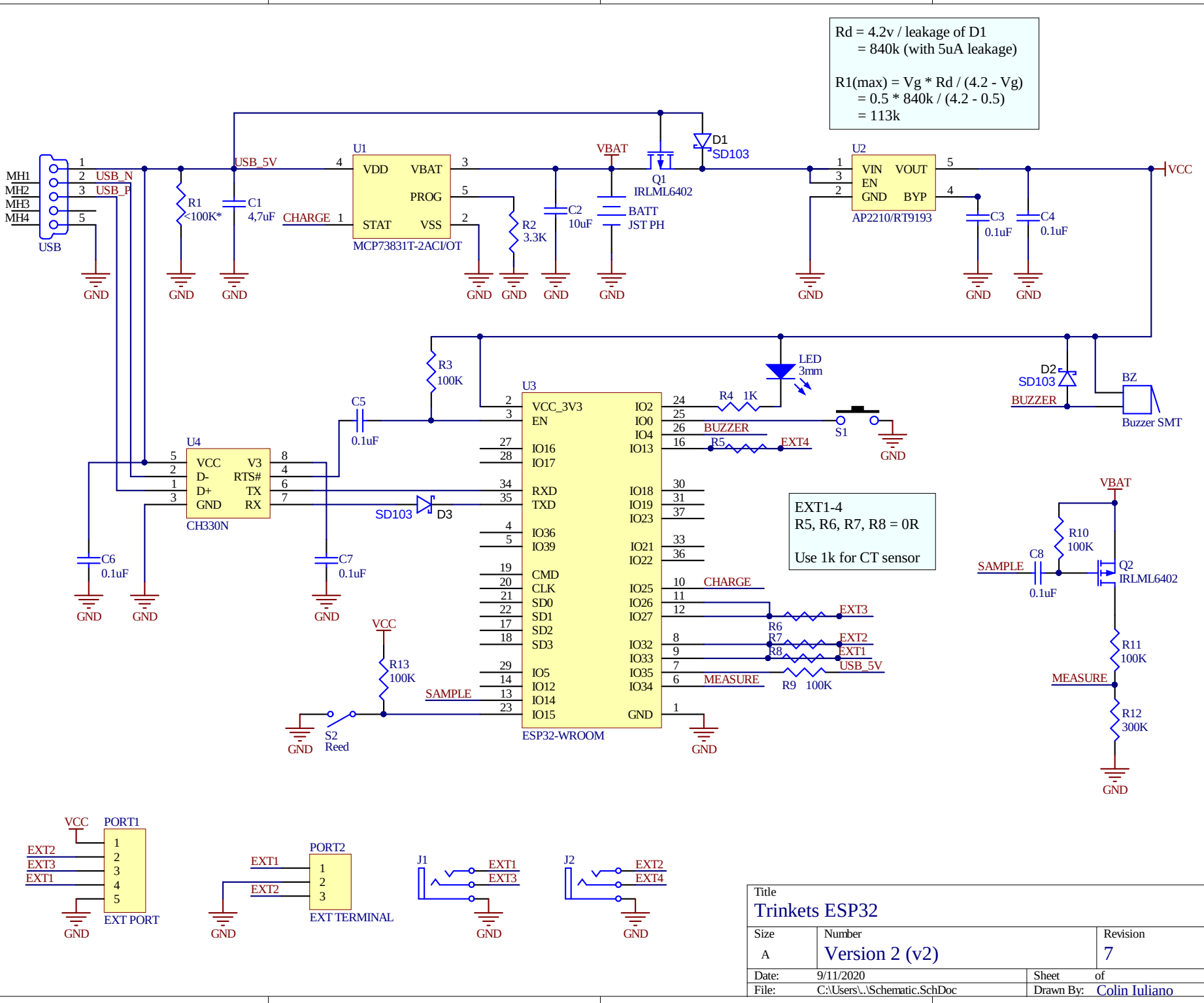

Rd = 4.2v / leakage of D1
= 840k (with 5uA leakage)
R1(max) = Vg * Rd / (4.2 - Vg)
= 0.5 * 840k / (4.2 - 0.5)
= 113k
USB
MH1
MH2
MH3
MH4
USB_N
USB_P
R1
<100K*
C1
4,7uF
USB_5V
U1
MCP73831T-2ACI/OT
VDD
VBAT
PROG
STAT
VSS
CHARGE
R2
3.3K
C2
10uF
VBAT
Q1
IRLML6402
D1
SD103
BATT
JST PH
U2
AP2210/RT9193
VIN
VOUT
EN
GND
BYP
C3
0.1uF
C4
0.1uF
VCC
R3
100K
LED
3mm
D2
SD103
BUZZER
BZ
Buzzer SMT
U3
ESP32-WROOM
VCC_3V3
EN
IO16
IO17
RXD
TXD
IO36
IO39
CMD
CLK
SD0
SD1
SD2
SD3
IO5
IO12
IO14
IO15
IO2
IO0
IO4
IO13
IO18
IO19
IO23
IO21
IO22
IO25
IO26
IO27
IO32
IO33
IO35
IO34
R4 1K
BUZZER
R5
EXT4
S1
C5
0.1uF
U4
CH330N
VCC
D-
D+
GND
V3
RTS#
TX
RX
SD103
D3
C6
0.1uF
C7
0.1uF
EXT1-4
R5, R6, R7, R8 = 0R
Use 1k for CT sensor
CHARGE
EXT3
R6
EXT2
R7
EXT1
R8
USB_5V
MEASURE
R9 100K
R13
100K
SAMPLE
S2
Reed
VBAT
R10
100K
C8
0.1uF
Q2
IRLML6402
R11
100K
R12
300K
PORT1
EXT PORT
PORT2
EXT TERMINAL
J1
J2
Title
Trinkets ESP32
Size
A
Number
Version 2 (v2)
Revision
7
Date:
9/11/2020
Sheet
of
File:
C:\Users\..\Schematic.SchDoc
Drawn By:
Colin Iuliano